



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	PLASMA PHYSICS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	A strong foundation in classical mechanics, electromagnetism, quantum mechanics, and fluid dynamics is essential as prerequisites for the course in plasma physics.				
Course Summary	The course in plasma physics provides an in-depth exploration of the behavior, properties, and applications of ionized gases, encompassing fundamental theories, experimental techniques, and practical implications across various fields such as astrophysics, fusion research, and industrial plasma technologies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate an understanding of the basic principles of plasma physics, including plasma formation and properties	Understand	Conceptual Understanding	Examinations, Assignments
CO2	Apply fluid dynamics concepts to analyze the behavior of plasmas and the propagation of waves within them	Apply	Application	Problem Sets, Lab Reports
CO3	Analyze the equilibrium and stability of plasma systems using relevant theoretical models and mathematical techniques	Analyze	Application	Research Papers, Projects
CO4	Evaluate plasma behavior and interactions based on kinetic theory, considering particle distribution functions and collisional processes	Evaluate	Application	Presentations, Discussions
CO5	Explain the physical mechanisms underlying wave propagation and instabilities in plasmas, considering both linear and nonlinear effects	Understand	Conceptual Understanding	Written Reports, Essays
CO6	Synthesize knowledge of plasma physics theories and principles to propose solutions to complex plasma-related problems in various applications	Create	Synthesis	Capstone Projects, Oral Defenses
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	INTRODUCTION TO PLASMA PHYSICS		13	18
	1	Existence of plasma, Definition of Plasma	2	
	2	Debye shielding 1D and 3D, Criteria for plasma	3	

	3	Applications of Plasma Physics (in brief)	1	
	4	Single Particle motions -Uniform E & B fields	3	
	5	Nonuniform B field, Non uniform E field	2	
	6	Time varying E field, Adiabatic invariants and applications	2	
	Sections 1.1 to 1.7.7, 2.1 to 2.8.3 of Book 1			
II	PLASMA AS FLUIDS AND WAVES IN PLASMAS		17	28
	7	Introduction –The set of fluid equations, Maxwell’s equations	1	
	8	Fluid drifts perpendicular to B, Fluid drifts parallel to B	2	
	9	The plasma approximations, Waves in Plasma - Waves, Group velocity, Phase velocity	2	
	10	Plasma oscillations, Electron Plasma Waves, Sound waves, Ion waves	2	
	11	Validity of Plasma approximations	2	
	12	Comparison of ion and electron waves	2	
	13	Electrostatic electron oscillations parallel to B, Electrostatic ion waves perpendicular to B	2	
	14	The lower hybrid frequency, Electromagnetic waves with B ₀ , Cutoffs and Resonances, Electromagnetic waves parallel to B ₀	2	
	15	Hydromagnetic waves, Magnetosonic wave	2	
	Sections 3.1 to 3.6, 4.1 to 4.20 of Book 1			
III	EQUILIBRIUM AND STABILITY		10	14
	16	Hydro magnetic equilibrium, The concept of b	2	
	17	Diffusion of magnetic field into plasma	3	
	18	Classification of instability, Two stream instability, the gravitational instability	3	

	19	Resistive drift waves, the Weibel instability	2	
	Sections 6.1 to 6.8 of Book 1			
IV	KINETIC THEORY		8	10
	20	The meaning of $f(v)$, Equations of kinetic theory	2	
	21	Derivation of the fluid equations	2	
	22	Plasma oscillations and Landau damping	2	
	23	the meaning of Landau damping, Physical derivation of Landau damping, Ion Landau damping	2	
	Sections 7.1 to 7.6.1 of Book 1			
V	OPEN ENDED MODULE			
	INTRODUCTION TO CONTROLLED FUSION		12	

Books and References:

1. F. F. Chen, Introduction to Plasma Physics and Controlled Fusion (Book 1)

2. K.L. Goswami, Introduction to Plasma Physics – Central Book House, Calcutta (Book 2)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	0	0	0	0	2	2	0	0	0	0	1
CO 2	2	2	1	0	0	0	2	2	0	0	0	0	1
CO 3	3	3	3	1	0	0	2	2	0	0	0	0	1
CO 4	3	2	3	2	0	0	2	2	0	0	0	0	1
CO 5	2	0	0	0	0	0	2	2	0	0	0	0	1
CO 6	3	0	0	0	0	0	2	2	0	0	0	0	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓